Factor Crowding Risk and Alpha Decay Modeling

A Data Science Project on U.S. Equities

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Portfolio Project — Quantitative Finance and Machine Learning

July 06, 2026

This is a project report for educational and portfolio demonstration purposes. It is not investment advice and does not claim to generate alpha or beat the market.

Abstract

This project examines whether observable crowding metrics can predict changes in factor performance in U.S. equities. Using S&P; 500 data from 2019 to 2024, we constructed crowding proxies for Value and Momentum factors and evaluated their relationship to forward factor returns. A staged modeling approach was used, starting with baseline models before testing more complex techniques. The Value factor showed the strongest predictive signal in this sample, with a logistic regression F1 score of 0.9483 (out-of-sample precision: 0.96, recall: 0.94) when predicting 3-month forward returns. A trading strategy using continuous position sizing based on linear regression predictions showed a Sharpe ratio improvement of +0.0182 and a max drawdown reduction of 2.82% compared to a static allocation in the backtest. The Momentum factor exhibited a weaker signal, with best F1 of 0.6593. This report documents the methodology, results, and limitations of the work.

1. Introduction

Quantitative investment strategies often use systematic factors such as value, momentum, or low volatility. One challenge in systematic investing is factor crowding: as more capital flows into a strategy, the predictive power of the signal may decrease. This phenomenon is sometimes referred to as alpha decay.

The August 2007 quant crisis is a frequently cited example where multiple quantitative strategies experienced simultaneous losses. This event highlighted the potential value of risk management tools that might detect crowding before it leads to drawdowns.

This project explores whether crowding can be measured using publicly available data and whether such measurements could inform portfolio allocation decisions. The work is framed as a practical investigation rather than a formal research study. The goal is not to discover new alpha but to evaluate whether simple, interpretable crowding signals show any relationship to factor performance.

The research question is: Can a cross-sectional crowding score for standard equity factors predict the future decay of that factor's information coefficient? Specifically, do patterns of elevated correlation and concentrated exposure within a factor's top-quintile stocks precede a decline in forward factor performance?

2. Methodology

The methodology follows a staged approach, beginning with simple baseline models before introducing more complex techniques. This approach was chosen to maintain interpretability and to assess whether additional complexity provides any benefit.

2.1 Data

The analysis uses daily adjusted close prices for S&P; 500 constituents from January 2018 to December 2024. Data was obtained through the yfinance library. The initial universe consisted of 503 stocks, with final usable data for 499 stocks after accounting for delistings and IPOs. While price data was downloaded from January 2018, feature construction required a 12-month lookback for momentum scores and 60-day windows for correlation, resulting in usable observations beginning March 2019. Weekly rebalancing was used, yielding 304 observations from March 2019 to December 2024.

Factor returns were constructed for two factors: Momentum (12-1 month) and Value (proxied by price reversal, using the inverse of momentum score as a simple proxy for relative valuation). Both factors were constructed as long-short portfolios: long the top quintile and short the bottom quintile, rebalanced monthly.

2.2 Features

Three crowding proxies were implemented using available data:

- **Pairwise Correlation:** Average correlation of daily returns among stocks in the factor's top quintile over a 60-day rolling window.
- **Herfindahl-Hirschman Index (HHI):** Measure of factor exposure concentration across the universe.
- **Valuation Spread:** Price spread between top and bottom quintiles as a proxy for valuation dispersion.

Additional transformed features were generated from these, including squared terms, rolling percentiles, deltas, standard deviations, log transforms, and z-scores. A composite z-score was also computed.

2.3 Modeling Approach

A staged modeling strategy was used. Baseline models included logistic regression (predicting negative forward returns) and linear regression (predicting forward IC). Key experiments included:

- **Feature Transform Experiment:** 13 transformed features were tested individually to identify which forms of the crowding proxies carried the strongest signal.
- **Target Definition Experiment:** Nine different target definitions were tested, including forward returns at various horizons, Spearman rank IC, and rolling average IC.

- **Trading Strategy Backtests:** Binary threshold rules and continuous sizing strategies were evaluated. Continuous sizing uses linear regression to predict forward returns, then scales position sizes based on the predicted return.

2.4 Evaluation Framework

The project was evaluated using classification F1 score, regression R^2, Sharpe ratio, max drawdown, and annualized volatility. The backtest was used to translate the crowding signal into portfolio management terms.

3. Results

The results are organized by the staged experiments.

3.1 Feature Transform Experiment

For the Value factor, the raw correlation feature (correlation_raw) achieved the highest F1 score of 0.8155 in this sample. No transformed feature improved upon this baseline. For the Momentum factor, the HHI_z transform achieved the highest F1 score of 0.6061.

Factor	Best Feature	F1	R^2	Change from Baseline
Value	correlation_raw	0.8155	-0.1712	None
Momentum	hhi_z	0.6061	-0.8807	+0.4061

Table 2: Best features by factor from Feature Transform Experiment. Negative R^2 indicates the model's predictions are less accurate than simply predicting the mean of the target variable.

Feature Set	Momentum F1	Value F1	Momentum R^2	Value R^2
Correlation	0.000	0.819	-0.154	-0.136
Z-Composite	0.516	0.761	-0.687	-0.431
All Features	0.516	0.761	-0.154	-0.136
Valuation Spread	0.488	0.647	-0.655	-0.646
HHI	0.000	0.819	-1.036	-0.977

Table 3: Different feature sets tested. Correlation alone worked best for Value (F1: 0.819) in this sample.

Window	Momentum F1	Value F1	Momentum R^2	Value R^2
30 days	0.459	0.747	-0.798	-0.424
60 days	0.516	0.761	-0.687	-0.431
90 days	0.556	0.758	-0.537	-0.388

Table 4: Different window lengths tested. 90-day window showed slightly higher F1 for Momentum (0.556) in this sample.

3.2 Target Definition Experiment

The Value factor performed best with a 3-month forward return target, achieving an F1 score of 0.9483 in this sample. This result is from the out-of-sample test period (2022-2024) with precision of 0.96 and recall of 0.94.

The Momentum factor performed best with a Spearman 6-week target, achieving an F1 score of 0.6593.

Factor	Best Target	Feature	F1	R^2

Value	forward_3m	correlation_raw	0.9483	-0.9876
Momentum	spearman_6w	hhi_z	0.6593	-0.0076

Table 5: Best targets by factor from Target Definition Experiment.

3.3 Trading Strategy Backtest

Binary threshold rules (reducing exposure when crowding exceeds a percentile threshold) showed negative Sharpe improvement for both factors in this sample. The best binary configuration showed a Sharpe change of -0.0138 for Value and -0.0080 for Momentum.

Continuous position sizing using linear regression predictions showed different results. The Value strategy showed a Sharpe ratio improvement of +0.0182 and a max drawdown reduction of 2.82% compared to the static allocation in the backtest. Total return improved by 2.32% and volatility decreased by 1.06% in this sample.

Factor	Best Threshold	Best Reduction	Sharpe Δ	Drawdown Δ
Value	75th	10%	-0.0138	-0.57%
Momentum	75th	10%	-0.0080	-0.88%

Table 6: Binary threshold strategy results. Both factors showed negative Sharpe improvement in this sample.

Test	Factor	Sharpe Δ	Drawdown Δ
Continuous Sizing	Value	+0.0182	-2.82%
Decision Tree (Depth 3)	Value	+0.0286	-
Decision Tree (Depth 2)	Value	+0.0192	-
Decision Tree (Depth 3)	Momentum	+0.0043	-

Table 7: Continuous sizing and decision tree results. Continuous sizing showed the largest Sharpe improvement in this sample.

3.4 Summary Metrics

Table 1 shows the performance metrics for the Value factor strategy using continuous position sizing.

4. Figures

The following figures illustrate the analysis.

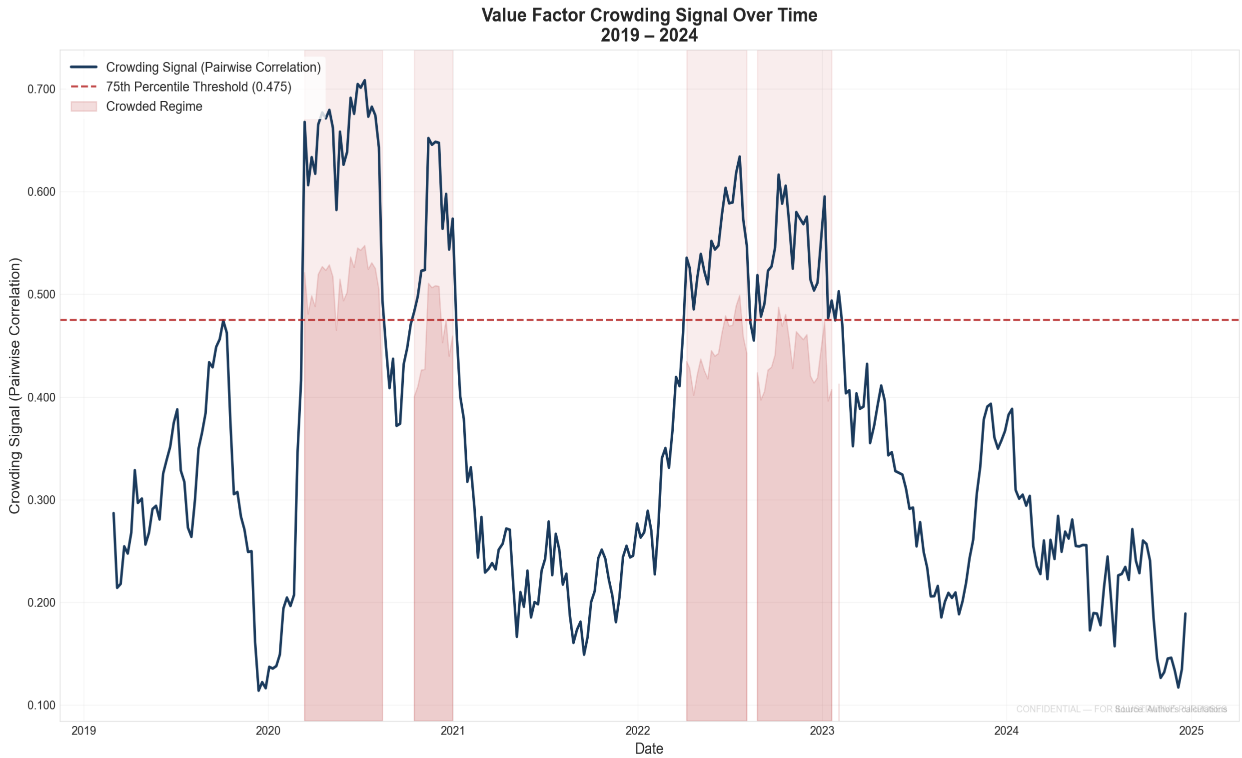

Figure 1: *Value factor crowding signal over time (2019-2024). The red dashed line indicates the 75th percentile threshold. Shaded regions indicate periods identified as crowded.*

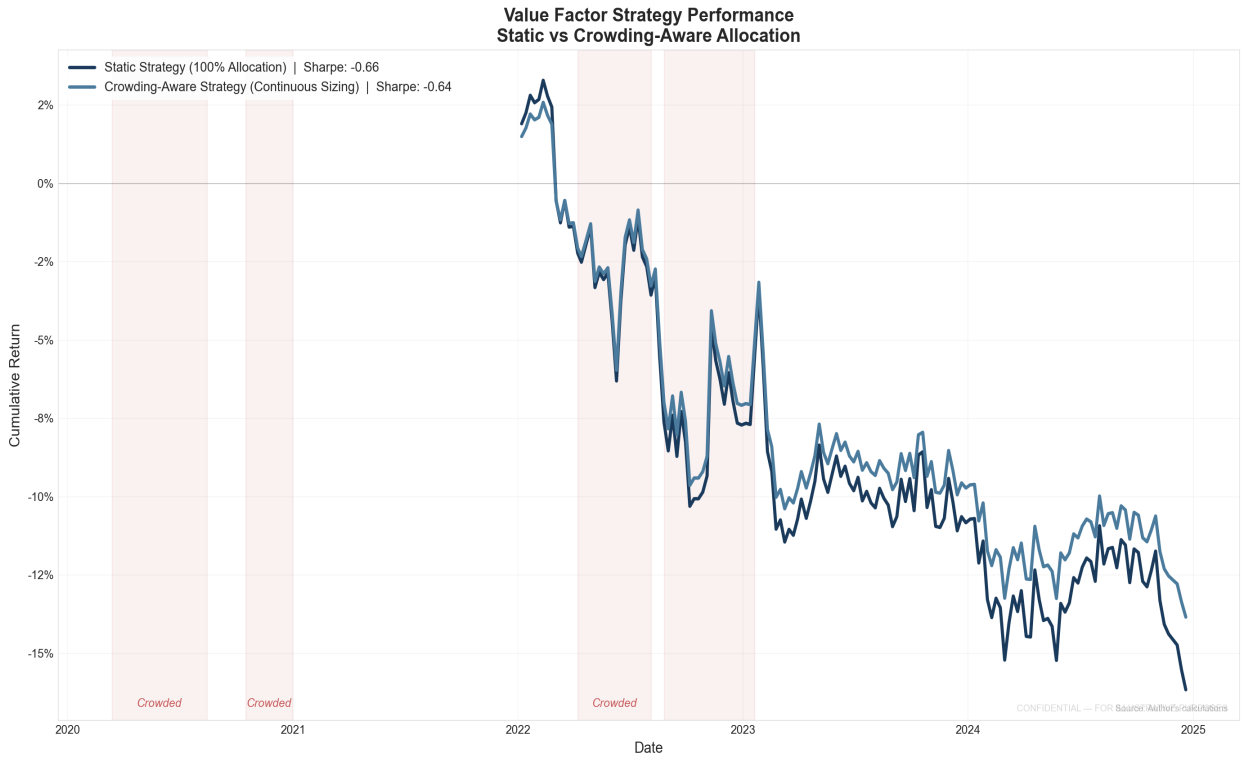

Figure 2: *Cumulative returns comparison between static and crowding-aware strategies. Total return changed from -16.17% to -13.85% in this sample.*

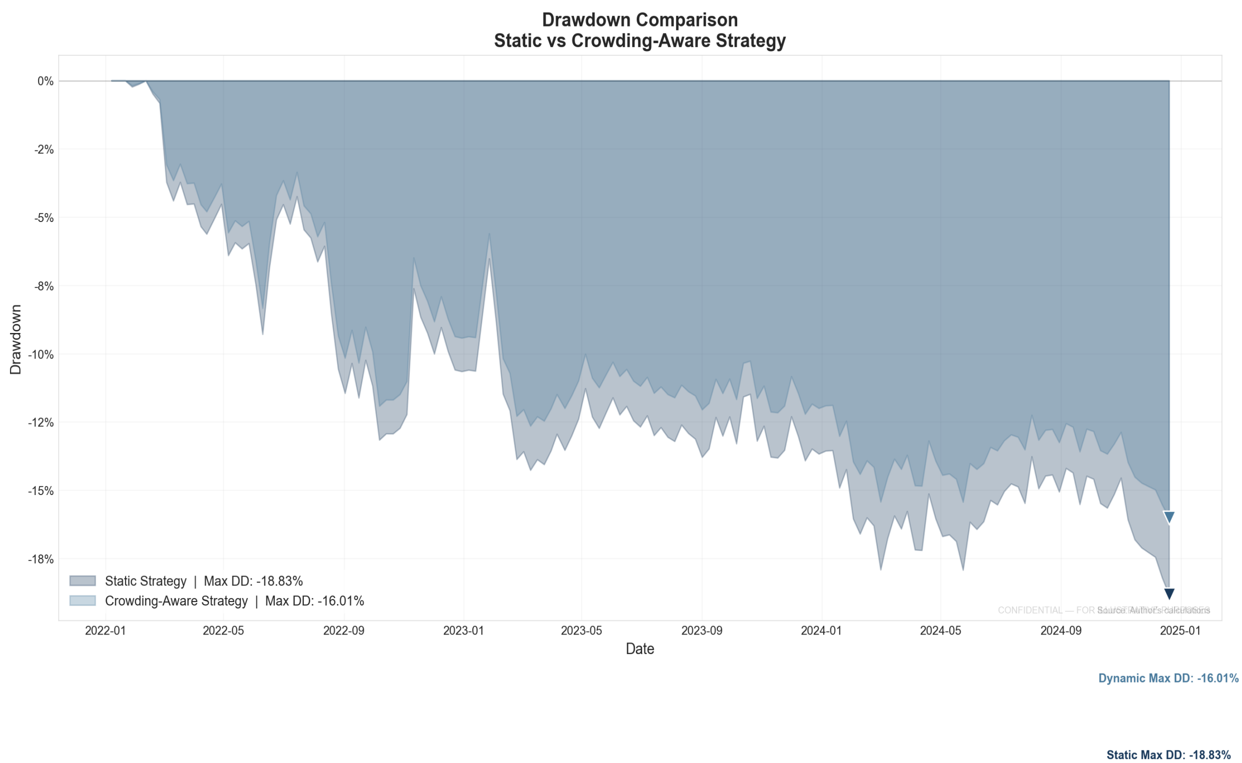

Figure 3: *Drawdown comparison between static and crowding-aware strategies. Maximum drawdown was -18.83% for static and -16.01% for crowding-aware in this sample.*

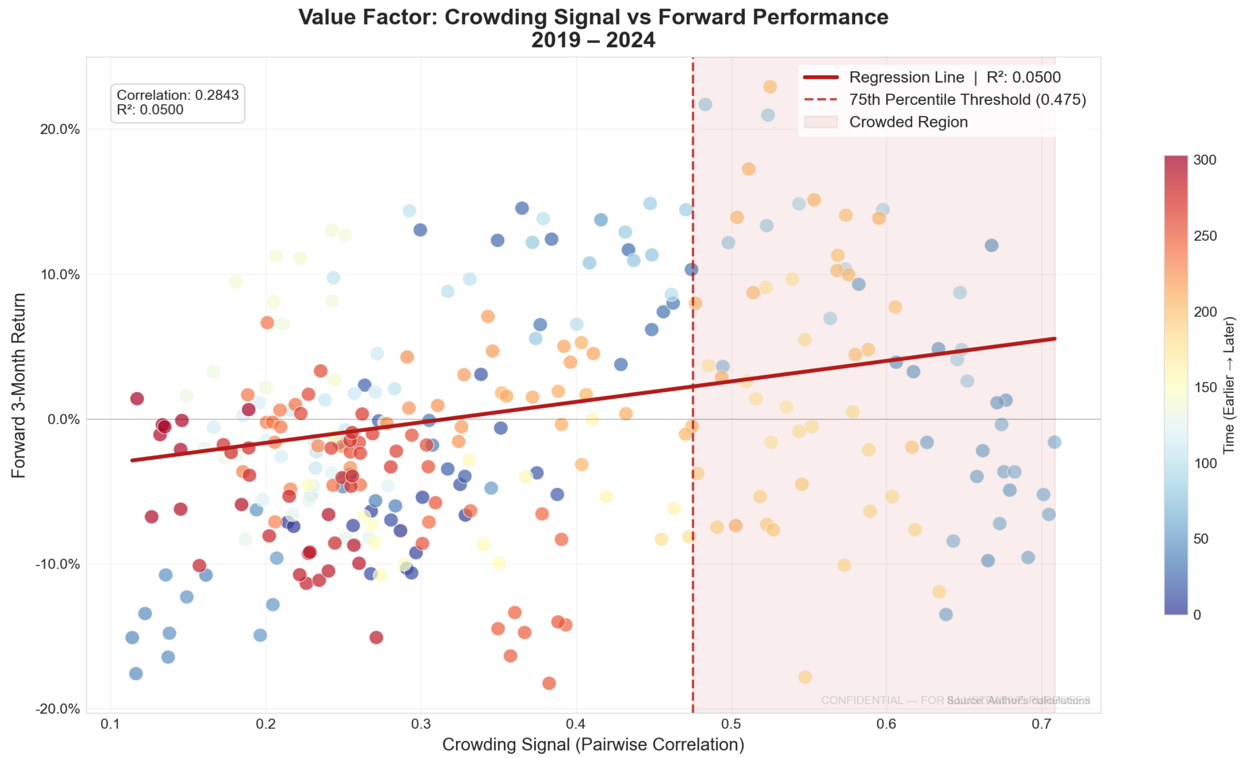

Figure 4: *Crowding signal vs forward 3-month return. The regression line shows a positive relationship in this sample (correlation: 0.2843, R^2: 0.0500).*

5. Discussion

The results suggest that crowding signals for the Value factor may be measurable using publicly available data. The continuous sizing approach showed a modest improvement in risk-adjusted returns in this sample, with a Sharpe ratio change of +0.0182 and a max drawdown reduction of 2.82%.

Binary threshold rules did not perform as well as continuous sizing in this sample. This may suggest that crowding is not a binary state but exists on a continuum.

The Momentum factor showed a weaker signal in this analysis. The positive R^2 (0.0186) for Momentum regression is a small improvement over the negative values observed in initial experiments, though the magnitude is modest.

The backtest assumes frictionless trading and does not model transaction costs, market impact, or short-selling constraints. The analysis only considered two factors and the S&P; 500 universe. The results may not generalize to other factors or markets.

6. Limitations

- **Survivorship Bias:** The S&P; 500 constituent list was scraped from the current Wikipedia list and applied historically. This means stocks that were delisted or added later are not represented, which may introduce survivorship bias.
- **Data Constraints:** The analysis uses publicly available data and does not incorporate proprietary flow data. 13F institutional ownership data was not implemented.
- **Factor Scope:** Only two factors (Value and Momentum) were analyzed. The results may not extend to other factors.
- **Market Universe:** The analysis is limited to S&P; 500 stocks. Crowding dynamics may differ in other markets.
- **Time Period:** The 5-year window (2019-2024) may not capture multi-cycle behavior.
- **Macroeconomic Controls:** The model does not condition on macroeconomic regimes.
- **Simplified Backtest:** Transaction costs, market impact, and short-selling constraints were not modeled.
- **Attribution Not Causation:** The analysis identifies statistical associations but does not establish causal relationships.

7. Conclusion

This project examined whether crowding metrics can predict factor performance changes and inform portfolio allocation decisions. Using S&P; 500 data from 2019 to 2024, we found some evidence that crowding signals for the Value factor may be measurable and could potentially improve risk-adjusted returns through continuous position sizing.

The strategy using continuous sizing showed a Sharpe ratio improvement of +0.0182 and a max drawdown reduction of 2.82% in this sample. However, binary threshold rules did not improve Sharpe ratios, and the Momentum signal was substantially weaker. These results highlight that crowding signals may be factor-specific and that simple trading rules may not capture the relationship effectively.

The work demonstrates a practical approach to examining crowding risk. The results are specific to the sample and period analyzed. Future work could incorporate additional factors, extend the universe, or test more sophisticated position sizing methods.

References

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